Continental
The Future in Motion

Case Study – Data Engineering

Scope

You are expected to perform **two main tasks**:

1. Case Study
   - Composed of **three sub-tasks**
2. Code Snippet
   - Composed of **one sub-task**

Each sub-task will be introduced in the following slides and identified using the icon.
Feel free to use the format you find most appropriate to present.

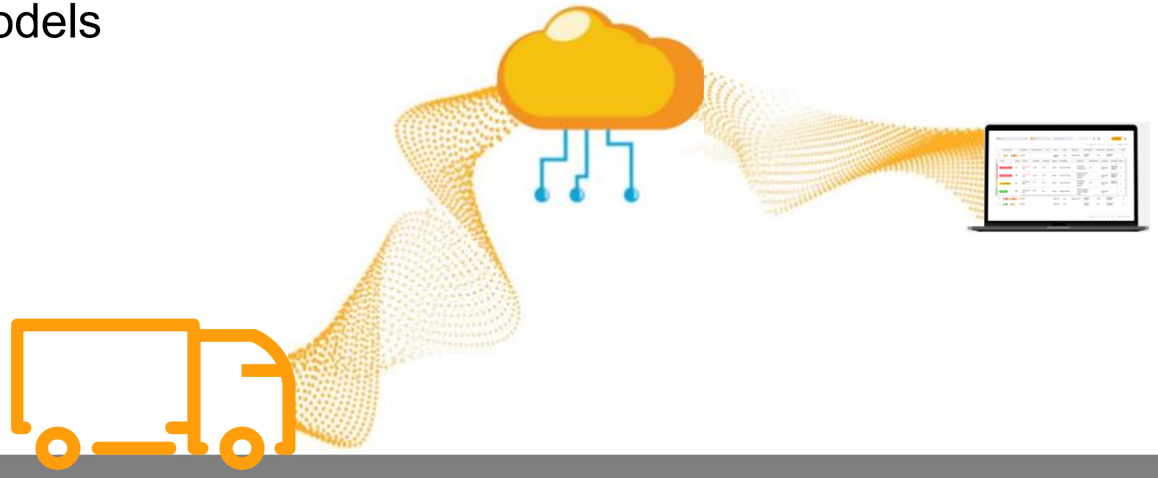
Case Study: Fleet telemetry data ingestion

Your team is responsible for ingesting **telemetry data from tires mounted on vehicles**.

Each **tire is equipped with dedicated sensor which periodically sends mileage readings** via a streaming pipeline, which are landed daily as Parquet files.

The business relies on this data to:

- Understand fleet utilization
- Detect inactive vehicles
- Identify data quality issues early
- Support downstream analytics, reporting and ML models



Case Study

Task

As a **Data Engineer**, you are asked to **design and implement an ingestion and transformation pipeline** for vehicle mileage telemetry data.

In addition to building the pipeline, you are expected to **answer a set of analytical and data quality questions** based on the processed data.

The telemetry data to be ingested is provided in the following Parquet files:

data_for_mileage_ingestion.parquet
mapping_vehicle_tire.parquet

You are expected to:

1. Design an efficient data ingestion, apply transformations and **enforce data quality rules** to prepare the data for analytical use. (Slide 6)
2. Support analytical questions from Business perspectives. (Slide 7)
3. Propose a suitable data storage design for downstream consumption. (Slide 8)

Case Study
Dataset

Main characteristics:

Event-based data (multiple records per tire per day)
Possible **duplicates**
Possible **out-of-order events**
Mileage should be **cumulative**
Mileage is sensor based, meaning it is **susceptible to hardware errors**.
One tire belongs to **one vehicle at a time**
Vehicles may have multiple tires
Mount date is optional field that is input manual from user

Files:

```
data_for_mileage_ingestion.parquet      #Contains telemetry events of mileage for tires mounted on vehicles.
mapping_vehicle_tire.parquet            # Provides the relationship between tires and vehicles.
```

Case Study
Data Ingestion

Please implement the requirements and answer the Data Engineer questions.

Requirements:

Filter:
- *It is necessary to ingest only data for September*

Quality rules: (Also think about extra rules that should be implemented)
- *No duplicates records*
- *Mileage must not decrease for the same tire over time.*

Processing:
- *The ingested mileage on the system should be latest known mileage for each tire on each day.*

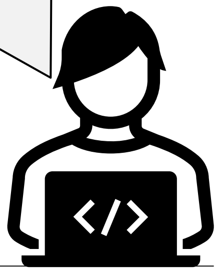
Q1

How many rows were ingested initially for September?

How many rows remain after filtering / cleaning / processing?

How many rows violated data quality rules?

How would you implement and monitor the pipeline?



Hint: Be mindful of physical plausibility of data

Case Study
Business perspective

We have telemetry data coming from tires mounted on vehicles.

Each event contains mileage information, timestamps, and identifiers for tires.

The business wants to understand **usage patterns**, **high runners**, and **inactive vehicles** for September on their fleet, please also address the following questions.

Q2

Which tires are the top tires (top 10) with more drove mileage (high runners) , what is their total mileage and total number of rows for September?

Which vehicles run more kilometers (Top 10)? and how much did they run?

Which vehicles remained inactive for more than 7 consecutive days, and for how long were they inactive?

Case Study

Part 2 - Data Modeling

Based on the provided input files:

data_for_mileage_ingestion.parquet — event-level mileage telemetry reported by tires

mapping_vehicle_tire.parquet — mapping between tires and vehicles

Q3 You are asked to **design a conceptual data warehouse model** that supports both **historical analysis** and **operational reprocessing**.

Please clearly outline the **tables and their principal columns, including** any business and technical attributes you consider necessary.

The warehouse must serve analytical use cases such as fleet utilization, tire lifecycle tracking, and daily mileage reporting, while remaining **scalable, auditable, and safe to rerun**.

Required Dimensions
The warehouse model should include (at minimum):

Tire dimension

- tire identifier
- article information
- position
- season (e.g., summer/winter/all-season)
- lifecycle attributes

Vehicle dimension

- vehicle identifier
- static vehicle attributes

Article / Product dimension

- article number
- product characteristics

Tires can be mounted on multiple vehicles over time.

Hint: Consider incorporating control and technical columns to support data quality, lineage, and reprocessing.

Code Snippet

Please share and explain an **illustrative code sample** from a previous project you developed.

In your explanation, cover the following:

- Why you selected this specific piece of code
- The context in which it was developed
- The problem or task the code is designed to solve

Thank you